

Main Terms in Operating System (OS)

1. Task Manager

- **What it is:** A system utility that shows you all the programs, processes, and services currently running on your computer.
- **Shortcut:** Ctrl + Shift + Esc or Ctrl + Alt + Del
- **Main Uses:**
 - See which app is using CPU, RAM, Disk, Network
 - End a non-responding program
 - Monitor system performance

 **Example:** If your system hangs or becomes slow, open Task Manager to check which app is using high CPU or memory.

2. Registry (Windows Registry)

- **What it is:** A database that stores all settings and configuration for Windows and installed software.
- **Tool:** regedit.exe
- **Structure:** Like folders, with keys and values (tree format)
- **Caution:** Modifying it wrongly can damage your system.

 **Example:** It stores settings like default browser, startup apps, system policies, etc.

3. File System

- **What it is:** A method the OS uses to store and organize files on storage devices (HDD/SSD).
- **Common Types:**
 - NTFS (Windows)
 - FAT32 (USB drives)
 - ext4 (Linux)
- **Functions:**
 - Track file location
 - Provide access permissions

4. Process and Thread

- **Process:** A program that is running (e.g., Chrome.exe)
- **Thread:** A smaller part inside a process that performs tasks.
- **Multithreading:** Running multiple threads at once for faster performance.

 **Example:** A browser is a process; each tab can be a thread.

5. 🚦 Scheduler

- **What it is:** Part of OS that decides **which process** runs **at what time**.
- **Goal:** Maximize efficiency and fairness.

🔍 **Example:** While you're listening to music and downloading a file, scheduler switches tasks so both run smoothly.

6. 🛡️ User Account Control (UAC)

- **What it is:** A Windows security feature that asks permission before allowing apps to make system-level changes.
- **Popup example:** "Do you want to allow this app to make changes to your device?"



7. 🛠️ System Services

- **What it is:** Background programs that run without user interaction.
- **Examples:**
 - Windows Update
 - Print Spooler
 - Antivirus services

8. ⚙️ Control Panel / Settings

- **What it is:** A GUI (graphical interface) to change system settings like network, sound, display, user accounts, etc.

9. 🌀 Boot Manager / Boot Loader

- **What it is:** Helps load the operating system when you power on the computer.
- **Examples:**
 - Windows Boot Manager

- GRUB (in Linux)

10. Kernel

- **What it is:** The **core part** of the OS.
- **Responsibilities:**
 - Communication between hardware and software
 - Process management
 - Memory management
 - Device control

✓ . Task Scheduler

- A tool in Windows that **automatically runs tasks** (like opening apps or backups) at set times or events.
- Example: Run antivirus every night at 10 PM.

✓ . AppData Folder

- A hidden folder that stores **user-specific data** like settings, history, and cache for apps.
- Located at: C:\Users\YourName\AppData
- It has three parts:
 - **Local** – for temporary files.
 - **Roaming** – for data that can sync (like browser settings).
 - **LocalLow** – for low-permission apps.

✓ . Task Manager

- A system monitor tool that shows:
 - **Running apps and processes**
 - **CPU, RAM, Disk, Network usage**
 - **Startup apps**
- Shortcut: Ctrl + Shift + Esc

✓ . Services

- Programs that run in the **background** to support Windows functions.
- Example: Windows Update service.
- Access: services.msc

✓ . Background Process

- Any task running **without user interaction**.
- Example: Google Drive sync, antivirus scan.

- Managed through Task Manager.

✓ . DLL Files (Dynamic Link Library)

- Files that store **shared functions/code** used by multiple programs.
- Example: Instead of writing print function in every app, they all use a common DLL.

✓ . EXE Files (Executable Files)

- Files used to **run software**.
- Double-clicking an .exe file starts the program.
- Example: word.exe opens MS Word.